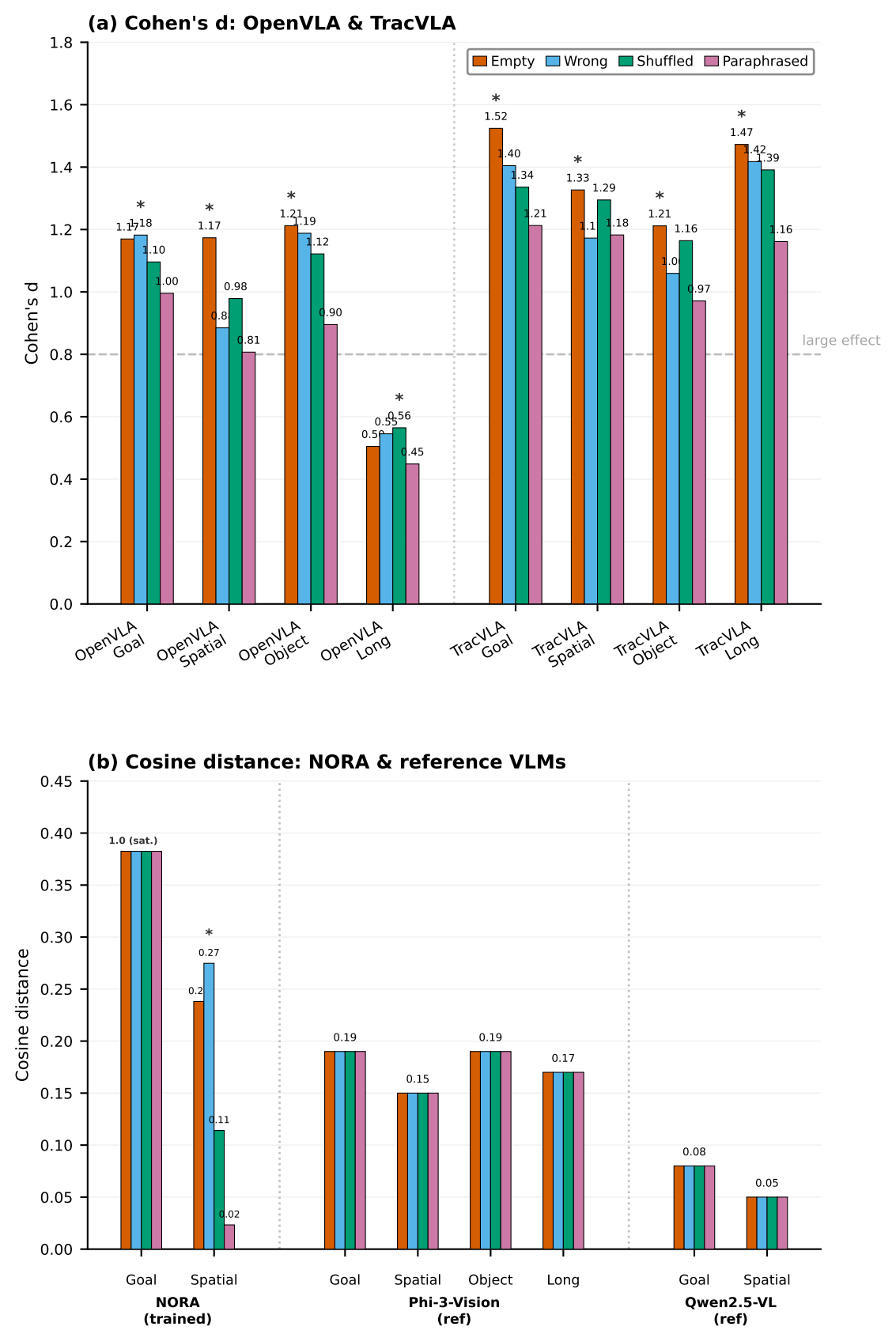
(a) Cohen's d: OpenVLA & TracVLA
Empty
Wrong
Shuffled
Paraphrased
Cohen's d
large effect
OpenVLA Goal
OpenVLA Spatial
OpenVLA Object
OpenVLA Long
TracVLA Goal
TracVLA Spatial
TracVLA Object
TracVLA Long
(b) Cosine distance: NORA & reference VLMs
Cosine distance
1.0 (sat.)
Goal
Spatial
NORA (trained)
Object
Long
Phi-3-Vision (ref)
Qwen2.5-VL (ref)